INCH-POUND

MIL-P-49151C(CR)
26 May 1992
SUPERSEDING
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8 November 1989

MILITARY SPECIFICATION

POWER ADAPTER ASSEMBLY
VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications - Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Power Adapter Assembly which is part of Viewer, Driver's Night Vision, AN/VVS-2(V).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.d).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TM Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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SPECIFICATIONS

MILITARY

MIL-P-11268	-	Parts, Materials, and Processes Used in Electronics Equipment
MIL-M-13231	-	Marking of Electronics Equipment Items

STANDARDS

MILITARY

MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700, Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.d).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-771200 - Adapter Assembly, Electrical Power

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The power adapter herein is referred to as the adapter assembly. The adapter assembly is an electrical component on the Driver's Viewer which converts available tracked vehicle voltage of 22 to 32Vdc to 2.7Vdc.

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3.2 Construction. The adapter assembly shall be constructed in accordance with SM-D-771200 and as specified herein.

3.3 Qualification. Unless otherwise specified by the contractor or purchase order, Power Adapter Assemblies furnished under this specification shall be products approved as qualified products (see 3.10.3 and 6.3).

3.3.1 Initial production testing. Supplier shall furnish and test the adapter assembly as specified in the contract of purchase order (see 6.2.c).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Performance characteristics.

3.5.1 Regulation. The adapter assembly shall regulate the output to 2.7, ± 0.1 volts direct current (Vdc) for any input voltage from 22Vdc to 32Vdc.

3.6 Environmental characteristics.

3.6.1 High temperature. The adapter assembly shall operate at a temperature up to +52°C. The assembly shall withstand storage temperature of +68°C without damage or degradation of performance (see 3.10.1 and 3.10.5).

3.6.2 Low temperature. The adapter assembly shall operate at a temperature down to -51°C. The assembly shall withstand storage temperature of -51°C without damage or degradation (see 3.10.1 and 3.10.5).

3.6.3 Temperature shock. The adapter assembly shall withstand cycles of sudden changes in temperature between -51°C and +68°C without damage or degradation of performance (see 3.10.1 and 3.10.5).

3.6.4 Humidity. The adapter assembly shall withstand +68°C and 94, ± 4 percent humidity without damage or degradation of performance see 3.10.1 and 3.10.5).

3.6.5 Vibration. The adapter assembly shall withstand vibration equivalent to that encountered in tracked vehicles, during 5000 miles of operation without damage or degradation of performance (see 3.10.1 and 3.10.5).

3.6.6 Shock.

3.6.6.1 Basic shock. The adapter assembly shall withstand a basic shock of 40g's (see 3.10.6) at 18 milliseconds duration without damage or degradation of performance (see 3.10.1 and 3.10.5).

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3.6.6.2 Weapon shock. The adapter assembly shall withstand a simulated weapon firing shock of 100 g's at 1.5 milliseconds duration, one half sine pulse, without damage or degradation of performance (see 3.10.1 and 3.10.5).

3.7 Treatment and painting. Unless otherwise specified, treatments and painting shall conform to MIL-STD-171.

3.8 Marking. Unless otherwise specified, the adapter assembly shall be marked and identified per MIL-M-13231.

3.9 Workmanship. Standards of workmanship shall be such that components will meet all requirements of this specification and any referenced specifications or drawings where inspected in accordance with Section 4. Parts and assemblies shall comply with requirement 9 of MIL-STD-454.

3.10 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.10.1 Damage. Damage is defined as any breakage, loosening, shift, dents. Failure of any finish, hardware, connection, or component. Moisture on any internal surface.

3.10.2 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

3.10.3 Qualified product. A product that has successfully met all the requirements of an IPT or First Article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

3.10.4 Environmental tests. For the purpose of this specification, environmental tests are those tests discussed in 4.6.

3.10.5 Degradation of performance. Any deterioration in quality, or failure to operate in accordance with the requirements of this specification.

3.10.6 "g". "g" is a unitless measure of the quantity of units of force associated with the earths gravity that a object is subjected to.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. Inspections shall be classified as follows:

a. Initial production testing (4.3).
b. Quality conformance inspection (4.4).
c. Inspection for packaging (4.7).

4.3 Initial production testing.

4.3.1 Inspection. Each IPT adapter assembly shall be inspected in accordance with Table I, either in process or when assembled, as required. Presence of one or more defects shall be cause for rejection of that assembly and may be cause for rejection of IPT. Unless otherwise specified (see 6.2.c) ten adapter assemblies manufactured with production tooling shall be selected from the first production lot and provided for IPT.

TABLE I. Inspection.

Inspection	Requirement paragraph
Any part or materials not as specified	3.2, 3.4
Treatment and painting not as specified	3.7
Markings not as specified or illegible	3.8
Workmanship	3.9

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4.3.2 IPT tests. Following successful completion of the inspections specified in 4.3.1, each IPT adapter assembly shall be subjected to all the test of Table II and may be performed in any order. Failure of any test shall be cause for rejection of the IPT.

TABLE II. Initial production testing.

Inspection	Requirement paragraph	Test paragraph
Regulation	3.5.1	4.5.1
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Humidity	3.6.4	4.6.4
Vibration	3.6.5	4.6.5
Shock	3.6.6	4.6.6

4.3.3 Disposition of IPT samples. Disposition of IPT samples shall be in accordance with the contract or purchase order (see 6.2.c).

4.4 Quality conformance inspection.

4.4.1 Examination. Each adapter assembly shall be examined as specified in Table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Test.

4.4.2.1 Group A inspection. Adapter assemblies that have passed the examination specified in 4.4.1, shall be tested in accordance with Table III. Failure of the test shall be cause for rejection of that unit. Adapter assemblies which pass Group A inspection shall be formed into lots by the contractor.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Regulation	3.5.1	4.5.1

4.4.2.2 Group B inspection. None required for this specification.

4.4.2.3 Group C inspection. This inspection shall consist of the tests specified in Table IV and shall be performed on adapter assemblies from lots which have passed Group A inspection. Sampling shall be in accordance with the contract or purchase order (see 6.2.e).

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4.4.2.4 Disposition of Group C samples. Item which have passed Group C tests and are not damaged (see 3.10.1) may be shipped against the contract.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Vibration	3.6.5	4.6.5
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Shock	3.6.6	4.6.6
Subgroup 2		
Humidity	3.6.4	4.6.4

4.4.2.5 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or order (see 6.2.b).

4.5 Test procedure. Unless otherwise specified all tests shall be performed at ambient temperature (see 3.10.2). Temperature chambers shall maintain temperatures within ± 2.0°C.

4.5.1 Regulation. Energize the adapter assembly with a variable power supply with a 51, ± 2 ohm load placed across pins 1 and 4 of the image tube connector. Vary the input voltage from 22Vdc to 32Vdc and measure the output of the adapter assembly. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.6 Operation and environment. The adapter assembly shall be retested to meet the requirements of 3.5.1 whenever an operating test is required during an environmental test, and at the completion of each environmental test (see 3.10.4). Failure to meet these requirements shall constitute failure as defined in degradation of performance (see 3.10.5).

4.6.1 High temperature. The adapter assembly shall be subjected to the test of Method 501.1, Procedure I, of MIL-STD-810, except that the temperature of Steps 4 and 5 (highest operating temperature) shall be +52°C and the highest storage temperature of steps 2 and 3 shall be +68°C. Steps 4 and 6 stabilization period shall be a minimum of two hours, and Step 5 operating test shall consist of the test of 4.5.1. Failure to meet the requirements of 3.5.1 and 3.6.1 shall constitute failure of this test.

4.6.2 Low temperature. The adapter assembly shall be subjected to the test of Method 502.1, Procedure I, of MIL-STD-810, except that the storage temperature of Step 2 shall be -51°C. Steps 2, 4, and 6 stabilization period shall be a minimum of two hours, and Step 5 operating test shall consist of the test of 4.5.1. Low operating temperature (Step 4) shall be -51°C. Failure to meet the requirements of 3.5.1 and 3.6.2 shall constitute failure of this test.

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4.6.3 Temperature shock. The adapter assembly shall be subjected to the test Method 503.1, Procedure I, of MIL-STD-810, except the high temperature chamber shall be maintained at +68°C, the low temperature chamber shall be maintained at -51°C, and Steps 1, 3, and 5 stabilization period shall be a minimum of two hours. Failure to meet the requirements of 3.5.1 and 3.6.3 shall constitute failure of this test.

4.6.4 Humidity. The adapter assembly shall be subjected to the test of Method 507.1, Procedure II, of MIL-STD-810C. Failure to meet requirements of 3.5.1 and 3.6.4 shall constitute failure of this test.

4.6.5 Vibration. The adapter assembly shall be subjected to MIL-STD-810, Method 514.2, Procedure VIII, Table 514.2-6, Tracked Vehicles, limited to one sweep in the vertical axis only. Failure to meet requirements of 3.5.1 and 3.6.5 shall constitute failure of this test.

4.6.6 Shock. The adapter assembly shall be subjected to the test of MIL-STD-810, Method 516.2.

a. 40 g's at 18 milliseconds duration Figure 516.2-1, shock procedure I.
b. Weapon firings, 100 g's using procedure IV except at 1.5 millisecond (ms) duration, Figure 516.2-2.

Failure to meet requirements of 3.5.1 and 3.6.6 shall constitute failure of this test.

4.7 Inspection for packaging. Inspection of packaging shall be in accordance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.g).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The adapter assembly is used with the AN/VVS-2(V) for driver's viewing during period of darkness.

6.2 Acquisition requirements. Acquisition documents must specify the following.

a. Title, number and date of this specification.
b. Necessary actions by contractor in the event of a Group C failure (see 4.4.2.5).
c. Quantity and schedule for Initial Production testing for all line items and disposition of IPT samples.

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d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
e. Sampling plans for Group C testing. As guidance, unless otherwise specified, sampling shall be conducted per the requirements of MIL-STD-105.
f. MIL-STD-810C is used for environmental tests of section 4.
g. Packaging requirements.

6.3 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.4 Definitions. See 3.10.

6.5 Subject term keyword listing.
Power Adapter Assembly
Viewer, Driver's

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - CR

Preparing Activity:
Army - CR

Project 5855-A362

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-49151C(CR)	26 May 1992

3. DOCUMENT TITLE
POWER ADAPTER ASSEMBLY VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
CECOM	(908) 532-5851	992-5851
c. ADDRESS (Include Zip Code) AMSEL-ED-TM Fort Monnouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340	

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